NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

Subsidiary Mathematics

033

09/07/2025 8.30 AM-11.30 AM

Names: .....

Index number: .....

SUBJECT: SUBSIDIARY MATHEMATICS

OPTION:
PHYSICS – CHEMISTRY - BIOLOGY **(PCB)**

DURATION: 3 HOURS

S6 QUESTIONS and ANSWERS BOOKLET

ADVANCED LEVEL NATIONAL EXAMINATIONS, 2024-2025

Instructions to candidates:

1) Write your names and index number on the answer booklet as written on your registration form.
2) DO NOT open this question paper until you are told to do so.
3) This paper consists of **two** sections: **A** and **B**.
   Section A: Attempt **all** questions. **(55 marks)**
   Section B: Attempt only **three** questions. **(45 marks)**
4) Geometrical instruments and silent non-programmable calculators may be used where it is necessary.
5) Use only a **blue** or **blac**k pen.

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

Questions	Marks	Questions	Marks
1		16	
2		17	
3		18	
4		19	
5		20	
6		21	
7		22	
8		23	
9		24	
10		25	
11		26	
12		27	
13		28	
14		29	
15		30	

Section A: Attempt all questions (55 marks).

For all multiple choice questions, circle a letter corresponding to the correct answer.

1) State whether each of the following statements is **True** or **False.** **(2 marks)**

   a) The union of set of rational and set of irrational numbers is the set of real numbers. .....

   b) The difference between set of real numbers and the set of positive real numbers is strictly negative integer numbers. .....

2) Which one among the following is the solution of the equation $4^{2x} = \frac{1}{64}$? Circle the right answer. **(2 marks)**

   a) -1
   b) 1
   c) $\frac{3}{2}$
   d) $-\frac{3}{2}$

3) Which answer among the ones given below corresponds to the $\cos(2\theta)$ if θ is an acute angle and that $\sin(\theta) = \frac{3}{4}$? Circle the right answer. **(2 marks)**

   a) $\frac{3}{4}$
   b) $\frac{1}{4}$
   c) $-\frac{1}{8}$
   d) $\frac{1}{8}$

4) The value of $\lim_{x \to 2}\left(\frac{x-2}{x^2-4}\right)$ is one among the following answers. Which one is the correct solution? Circle the right answer **(3 marks)**

a) $\frac{0}{0}$
b) ∞
c) 4
d) $\frac{1}{4}$

5) The derivative of this function $f(x) = \frac{-1}{x+1}$ at point $x = 3$ is one of the following answers. Which one? Circle the right answer. **(3 marks)**

a) $\frac{1}{16}$
b) $\frac{1}{4}$
c) $\frac{-1}{16}$
d) $\frac{-1}{4}$

6) Consider the parametric quadratic equation $x^2 - 6mx + 9 = 0$. Which is the value of the parameter m and all values of x among the ones given below, such that this equation admits a double positive real roots? Circle the right answer.

(4 marks)

a) $m = 1, x_1 = 3, x_2 = 3$

b) $m = -1, x_1 = -3, x_2 = -3$

c) $m = 1, x_1 = -3, x_2 = 3$

d) $m = 1, x_1 = 3, x_2 = -3$

7) Consider the matrix $A = \begin{pmatrix} 3 & -8 & 7 \\ 23 & 6 & -9 \\ 7 & 9 & 4 \end{pmatrix}$ and , $B = \begin{pmatrix} 4 & 50 & 60 \\ -7 & -5 & -12 \\ 4 & -13 & 16 \end{pmatrix}$. Choose and circle the answer that corresponds to $-A + 2B$. **(4 marks)**

a) $\begin{pmatrix} 5 & 108 & 113 \\ -37 & -16 & -15 \\ 1 & -35 & 28 \end{pmatrix}$

b) $\begin{pmatrix} 1 & -124 & 99 \\ 83 & 28 & -3 \\ 13 & 53 & -20 \end{pmatrix}$

c) $\begin{pmatrix} 1 & -124 & -99 \\ 83 & 28 & -3 \\ 13 & 53 & 20 \end{pmatrix}$

d) $\begin{pmatrix} 1 & -124 & -99 \\ 83 & 8 & -3 \\ 13 & 53 & -20 \end{pmatrix}$

8) The simple algebraic form of this complex number $i^{35}+i^{21}-i^{-1}+i^{-3}$ is one of the following answers, which one? Circle the right answer. **(4 marks)**

a) 0
b) $-2i$
c) 1
d) $2i$

9) Which answer among the ones given below is the sum of arithmetic terms $1+4+7+10+13+16+19+\ldots\ldots+58$? **(4 marks)**

a) 480
b) 560
c) 600
d) 590

10) The function g is given by $g(x)=ax^2-b$. If $g(2)=5$ *and* $g(-1)=2$. Which one among the following answers gives the value of a *and* b? Circle the right answer. **(3 marks)**

a) $a=1$ and $b=1$
b) $a=1$ and $b=-1$
c) $a=-1$ and $b=1$
d) $a=-1$ and $b=-1$

11) Given that $p(\mathrm{A})=\frac{1}{3}, p(\mathrm{B})=\frac{2}{5}$ and $p(\mathrm{A}\cup\mathrm{B})=t$. Which is the value of t among the ones below if A and B are Independent? Circle the right answer.

(5 marks)

a) $\frac{3}{5}$
b) $\frac{4}{5}$
c) 1
d) Does not exist

12) Which one among the answers given below is the solution of $(2\cos x-1)(3+2\cos x)=0$ in the interval $0\le x\le 2\pi$. Circle the right answer.

(4 marks)

a) $\frac{\pi}{3}$
b) $\frac{\pi}{3}, \frac{5\pi}{3}$
c) $\frac{\pi}{3}, \frac{5\pi}{3}, \cos^{-1}\left(\frac{-3}{2}\right)$
d) None of these

13) Consider the population p of a region where there is no immigration or emigration. The rate at which the population is growing is often proportional to the size of the population. This means larger populations grow faster, as we expect since there are more people to have babies. If the population has a continuous growth rate of 2% per unit of time, which among the answers given below corresponds to the population at time $t = 20 years$ if the initial population is 2 million? **(5 marks)**

a) $\approx 2983650.$

b) $\approx 2081622.$

c) $\approx 109196300.$

d) $\approx 2002001.$

14) Which one among the answers given below corresponds to the coefficient of correlation $r_{(x,y)}$ when $Cov(x,y) = -16.5$, $Var(x) = 2.89$, $Var(y) = 100$. **(5 marks)**

a) $r_{(x,y)} = 0.97$

b) $r_{(x,y)} = -0.97$

c) $r_{(x,y)} = -0.057$

d) $r_{(x,y)} = 0.057$

15) Find domain of the function $f(x)=\dfrac{x}{\sqrt{x^3-4x^2+x+6}}$ **(5 marks)**

Section B: Choose only three questions (45 marks).

Space in which to answer the three questions is at the end of section B.

16) a) Find equation of the tangent to the curve $y=x^2-4x-5$ at the point $(-1,8)$

(7 marks)

b) Find the value of $\int_6^8 \dfrac{1}{x^2-4x-5}dx$ **(8 marks)**

17) Solve by inverse matrix method the following simultaneous equation:

$$\begin{cases} 2x-y+5z=10 \\ x+y-3z=-2 \\ 2x+4y+z=1 \end{cases}$$

(15 marks)

18) a) Find the angle between the vector $\vec{u}=(7,1)$ and the vector $\vec{v}=(5,5)$

(5 marks)

b) Find the vector $\vec{w}=\vec{u}\times\vec{v}$ given $\vec{u}=(1,3,-3)$ and $\vec{v}=(4,3,1)$. Show that $\vec{u}$ and $\vec{w}$ are perpendicular. **(5 marks)**

c) Show that $\dfrac{\cos 2x-\sin 2x+1}{\cos 2x+\sin 2x-1}=\cot x$ whenever both expressions have meaning.

(5 marks)

19) a) Find the covariance between x and y if

x	3	4	5	6	7
y	8	7	6	5	4

(5 marks)

a) A bag (1) contains 4 red pens and 3 blue pens. Another bag (2) contains 3 red pens and 4 blue pens. A pen is taken from the first bag (1) and placed into the second bag (2). The second bag (2) is shaken and a pen is taken from it and placed in the first bag (1). If now a pen is taken from the first bag, use the tree diagram to find the probability that it is a red pen.

(10 marks)

20) An object of $5kg$ mass in motion is moving by a force $F(t)=75t+50$ Newton where t is the time.

a) Assuming that the force is given by the product of mass and its acceleration. Find its acceleration. **(2 marks)**

b) Find its velocity v as a function of time t, if $v(0)=5m/\sec$ **(7 marks)**

c) Find the displacement d as a function of time t if $d(0)=5\,\mathrm{m}$ **(6 marks)**

Space in which to answer the three chosen questions from section B.

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